Topological Spaces

(a) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\tau' = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}, \{a, b, d\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, d\} = \{a, b\} \in \tau'$

$\{a, b\} \cup \{a, b, d\} = \{a, b, d\} \in \tau'$, so τ' satisfies the axioms of a topology.

(b) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(c) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(d) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(e) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(f) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(g) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(h) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(i) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(j) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(k) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(l) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(m) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(n) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(o) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(p) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(q) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(r) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(s) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(t) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(u) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(v) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(w) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(x) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(y) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(z) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

Continuous Functions & Homeomorphisms

(a) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(b) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(c) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(d) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(e) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(f) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(g) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(h) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(i) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

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$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(j) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(k) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(l) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(m) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(n) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(o) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(p) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(q) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(r) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(s) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(t) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(u) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(v) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(w) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(x) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(y) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(z) $X = \{a, b, c, d\}$

$\tau = \{X, \emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$

$\{a, b\} \cap \{a, c\} = \{a\} \in \tau$

$\{a, b\} \cup \{a, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

$\{a, b\} \cap \{a, b, c\} = \{a, b\} \in \tau$

$\{a, b\} \cup \{a, b, c\} = \{a, b, c\} \in \tau$, so τ satisfies the axioms of a topology.

(a) $f(x) = x$ is continuous because $f^{-1}(U) = U \in \tau$ for all $U \in \tau$.

(b) $f(x) = x$ is continuous because $f^{-1}(U) = U \in \tau$ for all $U \in \tau$.

(c) $f(x) = x$ is continuous because $f^{-1}(U) = U \in \tau$ for all $U \in \tau$.

(d) $f(x) = x$ is continuous because $f^{-1}(U) = U \in \tau$ for all $U \in \tau$.

(e) $f(x) = x$ is continuous because $f^{-1}(U) = U \in \tau$ for all $U \in \tau$.

(f) $f(x) = x$ is continuous because $f^{-1}(U) = U \in \tau$ for all $U \in \tau$.

(g) $f(x) = x$ is continuous because $f^{-1}(U) = U \in \tau$ for all $U \in \tau$.

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(c) $f(x) = x$ is continuous because $f^{-1}(U) = U \in \tau$ for all $